



Diamagnetic response and phase stiffness for interacting isolated narrow bands

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Superconductivity is a macroscopic manifestation of a quantum phenomenon where pairs of electrons delocalize and develop phase coherence over a long distance. A long-standing quest has been to address the underlying microscopic mechanisms that fundamentally limit the superconducting transition temperature, T_c . A platform which serves as an ideal playground for realizing “high”-temperature superconductors are materials where the electrons’ kinetic energy is quenched and interactions provide the only energy scale in the problem. However, when the noninteracting bandwidth for a set of isolated bands is small compared to the interactions, the problem is inherently nonperturbative. In two spatial dimensions, T_c is controlled by superconducting phase stiffness. Here, we present a theoretical framework for computing the electromagnetic response for generic model Hamiltonians, which controls the maximum possible superconducting phase stiffness and thereby T_c , without resorting to any mean-field approximation. Our explicit computations demonstrate that the contribution to the phase stiffness arises from i) “integrating out” the remote bands that couple to the microscopic current operator and ii) the density–density interactions projected on to the isolated narrow bands. Our framework can be used to obtain an upper bound on the phase stiffness and relatedly T_c for a range of physically inspired models involving both topological and nontopological narrow bands with density–density interactions. We discuss a number of salient aspects of this formalism by applying it to a specific model of interacting flat bands and compare the upper bound against the known T_c from independent numerically exact computations.

unconventional superconductivity | bounds on superconductivity | nonperturbative methods

What is the highest attainable superconducting temperature T_c in a given physical system? One suggested theoretical route to enhance T_c has been to focus on the problem of “flat-band” superconductivity (1–3)—a strong to intermediate-coupling regime, where Bardeen–Cooper–Schrieffer (BCS) mean-field theory does not a priori apply. In particular, attempting to enhance the gap scale in the strong-coupling regime often comes at the cost of degrading the phase-coherence scale (4). However, the discovery of superconductivity across a number of two-dimensional (2D) moiré materials in the last few years (5–10) has refocused our attention on this question.

Across most of these platforms, the low-energy physics is associated with a set of partially filled “active” bands, which in the noninteracting limit have a narrow bandwidth, W . These active bands are well separated from the “remote” bands by a charge-gap, Δ , and the characteristic electronic interaction scale, V , is believed to be $W \lesssim V \ll \Delta$; Fig. 1. While the detailed microscopic mechanism for the origin of superconductivity across these materials remains unclear, there are basic conceptual and model-independent questions that have not been understood going beyond BCS theory. In two dimensions and in the limit of strong interactions, T_c is limited by the superconducting phase stiffness (11, 12), which is determined by the transverse electromagnetic response of the electronic system (13). One of the goals of this paper is to lay down the theoretical foundation for analyzing the low-energy electromagnetic response for a narrow-bandwidth system in the regime of strong interactions and computing the largest possible superconducting phase stiffness when interactions are projected to the isolated flat bands. An additional level of complexity arises from the nontrivial momentum dependence of the Bloch wavefunctions associated with the active bands, which affect the nature of interactions projected to these bands, as well as the effective current correlations obtained upon integrating out the higher-energy degrees of freedom. Characterizing the superconducting instabilities in this nonperturbative regime across their respective phase diagrams, and their detailed dependence on the various microscopic parameters, requires the development of new analytical and numerical methods.

Significance

In a series of recent breakthroughs, it has been demonstrated that two-dimensional (2D) moiré materials can support superconductivity in the limit of extremely narrow bands. Given the quantum mechanical wavefunctions and the form of interactions associated with the electrons, it is natural to address what is the maximum attainable superconducting transition temperature. In the absence of a microscopic theory of superconductivity in this strongly interacting regime, we lay down the framework for computing the maximum possible value of a fundamental characteristic, namely the superconducting phase stiffness, for interacting Hamiltonians that host narrow bands. The framework does not rely on the conventional Bardeen–Cooper–Schrieffer approximation and helps disentangle the role of different ingredients that enhance tendency toward superconductivity.

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The authors declare no competing interest.

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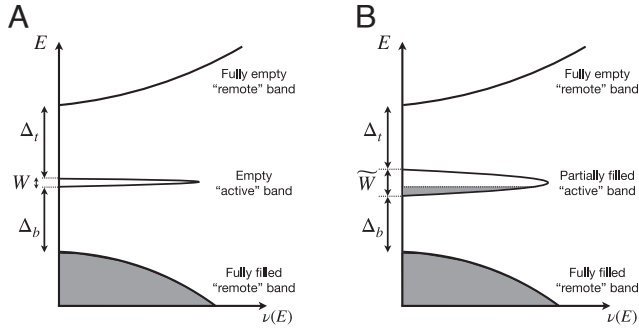


Fig. 1. A schematic of the energy bands in the (A) noninteracting and (B) interacting limits, respectively. In (A), we denote the bandwidth associated with the isolated narrow bands as W , which is well separated from the remote (Bottom and Top) bands by gaps, Δ_b , Δ_t , respectively. In (B), the interaction-induced renormalization of the isolated bands leads to a modified bandwidth, $\tilde{W}$. We assume a hierarchy of energy scales, $W \lesssim V(\sim \tilde{W}) \ll \Delta_{b,t}$. This manuscript derives the effective diamagnetic response, K_{xx}^{eff} , and the integrated optical spectral weight, $\int_0^\Lambda d\omega \text{Re}[\sigma_{xx}^{\text{eff}}(\omega)]$ (where $\tilde{W} < \Lambda < \Delta_t$), for the isolated bands with projected density-density interactions in the limit $\Delta \rightarrow \infty$.

Since a reliable microscopic theory of superconductivity for interacting flat bands is currently unavailable, an interesting point of view has been to formulate fundamental “bounds” on the superconducting phase-stiffness, $D_s(T)$, at a temperature, T . In two dimensions, the superconducting T_c and $D_s(T)$ are related to each other, $T_c = \pi D_s(T_c^-)/2$ (11). Within BCS mean-field theory (a priori unjustified) and in the superconducting ground state (i.e., ignoring competing instabilities), $D_s(0)$ is bounded from below by a geometric invariant (14–18), namely the Fubini–Study metric, associated with the Bloch wavefunctions (19). Within a set of restrictive conditions, the BCS wavefunction can be shown to be an exact ground state (20, 21), but $T_c = 0$ as a result of an emergent $SU(2)$ symmetry. However, numerically exact solution to the many-body problem in the presence of infinitesimal changes to the Hamiltonian is known to induce nonsuperconducting competing orders (22, 23), where $D_s(0) = 0$. On the other hand, given the entire bounded electronic spectrum (i.e., including all the bands) for any microscopic Hamiltonian with purely density–density interactions, the optical spectral weight integrated up to the full bandwidth serves as a rigorous upper bound on D_s (24). For most electronic solids, this will typically be $D_s \lesssim O(\text{eV})$. This bound is useful for Galilean-invariant systems (especially at low-densities), where T_c is bounded by the Fermi-energy, E_F (up to a numerical prefactor). More generally, there are no analogous bounds on T_c/E_F or on $T_c/D_s(0)$ in non-Galilean-invariant systems (25), which necessitates the development of a more widely applicable theoretical framework for addressing upper bounds on T_c .

Deriving a rigorous “low-energy” upper bound on T_c in the limit $\Delta \rightarrow \infty$, which is controlled by an interacting effective Hamiltonian, $\mathcal{H}_{\text{eff}}$, acting only on the active bands remains an outstanding challenge. This manuscript will be concerned with formulating the theoretical framework that is necessary to derive such an upper bound. This program depends on the strength and form (i.e., spatial profile) of the interactions. As a result, it will depend ultimately on the nature of the various multiparticle correlation functions within the low-energy manifold of states which is acted upon by $\mathcal{H}_{\text{eff}}$. A number of recent works have directly studied the many-body problem in sign-problem-free models using numerically exact quantum Monte-Carlo (QMC) computations (22, 23, 26–28). As a concrete example, we apply our framework to derive a theoretical upper bound on $D_s(0)$ for

a model of interacting flat bands below, where the exact result is known independently from QMC.

Model

In general, D_s can be computed as the transverse electromagnetic response (13, 29) for an effective Hamiltonian, $\mathcal{H}_{\text{eff}}[A]$, in the presence of a probe gauge-field, A ,

$$\frac{D_s}{\pi e^2} = \left[\langle K_{xx}^{\text{eff}} \rangle - \chi_{xx}^{\text{eff}}(\omega = q_x = 0, q_y \rightarrow 0) \right]_{A \rightarrow 0}, \quad [1a]$$

$$K_{xx}^{\text{eff}} = \frac{1}{2} \frac{\delta^2 \mathcal{H}_{\text{eff}}[A]}{\delta A_x \delta A_x}, \quad \chi_{xx}^{\text{eff}}(\mathbf{q}) = \langle J_x^{\text{eff}}(\mathbf{q}) J_x^{\text{eff}}(-\mathbf{q}) \rangle, \quad [1b]$$

where K_{xx}^{eff} is the effective diamagnetic contribution, and χ_{xx}^{eff} is the effective current susceptibility, with $J_x^{\text{eff}} = -\delta \mathcal{H}_{\text{eff}}[A]/\delta A_x$. The expectation values are evaluated with respect to the many-body state at inverse temperature $T^{-1} = \beta$ as $\langle \dots \rangle = \text{Tr}[e^{-\beta \mathcal{H}_{\text{eff}}} \dots] / \text{Tr}[e^{-\beta \mathcal{H}_{\text{eff}}}]$. One of the important contributions of our work will be explicit low-energy formulas for these effective operators and response functions, where we will highlight their key differences from the naive (microscopic) definitions for the same.

Given a UV Hamiltonian, H , which is distinct from $\mathcal{H}_{\text{eff}}$, there are a number of subtleties associated with computing these effective susceptibilities. To be concrete, we start with a generic translationally invariant UV Hamiltonian,

$$H = H_{\text{kin}} + H_{\text{int}}, \quad [2a]$$

$$H_{\text{kin}} = \sum_{\mathbf{r}, \mathbf{r}'} \sum_{\alpha, \alpha'} t_{\alpha\alpha'}(\mathbf{r} - \mathbf{r}') c_{\mathbf{r}\alpha}^\dagger c_{\mathbf{r}'\alpha'} - \mu N, \quad [2b]$$

$$H_{\text{int}} = \sum_{\mathbf{r}, \mathbf{r}'} V(\mathbf{r} - \mathbf{r}') n_{\mathbf{r}} n_{\mathbf{r}'}, \quad [2c]$$

where $c_{\mathbf{r}\alpha}$, $c_{\mathbf{r}\alpha}^\dagger$ denote microscopic electronic operators at site $\mathbf{r}$ with a collective orbital and spin index α , satisfying the standard anticommutation algebra. In H_{kin} , the matrix $t_{\alpha\alpha'}(\mathbf{r} - \mathbf{r}')$ includes a set of arbitrary intersite/orbital hoppings, and μ is an external chemical potential that couples to $N = \sum_{\mathbf{r}} n_{\mathbf{r}} = \sum_{\mathbf{r}, \alpha} c_{\mathbf{r}\alpha}^\dagger c_{\mathbf{r}\alpha}$. For simplicity, in the remainder of this work, we consider only density–density interactions, $V(\mathbf{r} - \mathbf{r}')$, at the UV scale (e.g., we ignore electron–phonon interactions, “correlated-hopping,” pair-hopping interactions, etc.). An upper bound on the phase stiffness for H can be evaluated—the probe gauge field couples only to H_{kin} , leading to an upper bound for D_s expressed in terms of the full optical spectral weight (24).

By transforming from the orbital to band basis, we obtain $H_{\text{kin}} = \sum_{\mathbf{k}, m} (\varepsilon_{\mathbf{k}m} - \mu) c_{\mathbf{k}m}^\dagger c_{\mathbf{k}m}$, where m represents (possibly degenerate) bands with Bloch functions, $|u_{\mathbf{k}, m}\rangle$. Then, let us recall that the microscopic current operator and the diamagnetic response can be obtained as ref. 13,

$$J_\mu(\mathbf{q}) = \sum_{\mathbf{k}, m, m'} c_{\mathbf{k}+\frac{\mathbf{q}}{2}, m}^\dagger c_{\mathbf{k}-\frac{\mathbf{q}}{2}, m'} \langle u_{\mathbf{k}+\frac{\mathbf{q}}{2}, m} | \partial_{k_\mu} \hat{h}_{\mathbf{k}} | u_{\mathbf{k}-\frac{\mathbf{q}}{2}, m'} \rangle, \\ K_{\mu\nu} = \sum_{\mathbf{k}, m, m'} c_{\mathbf{k}m}^\dagger c_{\mathbf{k}m'} \langle u_{\mathbf{k}, m} | \partial_{k_\mu} \partial_{k_\nu} \hat{h}_{\mathbf{k}} | u_{\mathbf{k}, m'} \rangle, \quad [3]$$

where $\hat{h}_{\mathbf{k}} = \sum_n (|u_{\mathbf{k}, n}\rangle \varepsilon_{\mathbf{k}n} \langle u_{\mathbf{k}, n}|)$ is the matrix of $(H_{\text{kin}} + \mu N)$.

For later convenience, note that the diamagnetic term and the current operator (in the small $\mathbf{q}$ limit) can also be written in the following way:

$$J_\mu(q_\mu \rightarrow 0) = -i[\hat{X}_\mu, H], \quad [4a]$$

$$K_{\mu\nu} = -[\hat{X}_\mu, [\hat{X}_\nu, H]], \quad [4b]$$

where $\hat{X}_\mu \equiv \sum_i x_i^\mu c_i^\dagger c_i$ is the many-body position operator.

For the remainder of our discussion, we will be interested only in the effective low-energy contribution to D_s from a finite subset, $\{m \in \text{active bands}\}$, that are separated from the remote bands by a large gap (Fig. 1). There are two conceptually equivalent approaches for addressing this problem: 1) In the limit where Δ is finite but large, the microscopic current operator couples together the active and remote bands; the effects of the remote bands have to be “integrated-out” via a Schrieffer–Wolff (SW) transformation (30). 2) The model has a global conserved $U(1)$ density associated with the total particle number. However, in the limit where $\Delta \rightarrow \infty$, there is an emergent conservation law associated with the electronic density restricted to the active bands and a corresponding current operator. The electromagnetic response can then be obtained by carrying out an appropriately defined gauge transformation in terms of “projected” coordinates, that does not take us out of the low-energy Hilbert space. One caveat is that by taking the second approach, one has to first determine how the physical electromagnetic response is related to the response to the emergent gauge transformation, which is a priori unknown. Within both approaches, it is worth noting that the projected interaction can generate a finite bandwidth, which in turn contributes to D_s .

We will begin by adopting the first approach, where we can analyze the effects of a finite Δ on the physics explicitly. Let us resolve the Hamiltonian into its “diagonal” and “off-diagonal” pieces, respectively,

$$H = H_d + H_o, \quad [5a]$$

$$H_d = \mathbb{P}H\mathbb{P} + \mathbb{Q}H\mathbb{Q}, \quad [5b]$$

$$H_o = \mathbb{P}H\mathbb{Q} + \mathbb{Q}H\mathbb{P}, \quad [5c]$$

where $\mathbb{Q} = \mathbb{I} - \mathbb{P}$. Here, $\mathbb{P}$ is the projection operator to the sub-Hilbert space, $\mathbb{H}$, spanned by the many-body states with partially occupied active bands and fully occupied (empty) lower energy (higher energy) remote bands; see Fig. 2 for a schematic depiction of the action of different projections. The basis of $\mathbb{H}$ with n occupied states in the active bands can be constructed from the “vacuum” state, $|\psi_0\rangle$, corresponding to fully filled remote bands as,

$$|\psi^{(n)}\rangle = \prod_{p=1}^n c_{\mathbf{k}_p m_p}^\dagger |\psi_0\rangle. \quad [6]$$

Here, the set of m_p belongs to the set of active bands, and the state $|\psi^{(n)}\rangle$ has total momentum, $\sum_p \mathbf{k}_p$. Since the total number of particles is conserved, any many-body state that does not belong in this low-energy manifold necessarily has holes (electrons) in the lower (upper) bands and is therefore well separated in energy. Moreover, later, when we consider the limit of $\Delta \rightarrow \infty$, there is an emergent conservation law associated with the density of electrons in the active bands. Finally, we note that the low-energy Hamiltonian of interest to us is $\mathcal{H}_{\text{eff}} \equiv \mathbb{P}H\mathbb{P} = \mathbb{P}H_d\mathbb{P}$.

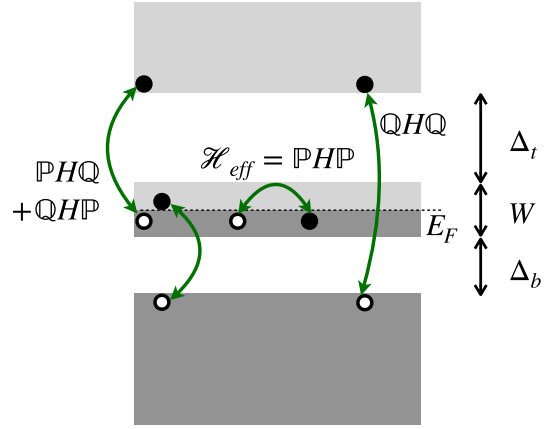


Fig. 2. A schematic depiction of the projections and Schrieffer–Wolff transformations. $\mathcal{H}_{\text{eff}}$ is the effective Hamiltonian which acts only within the low-energy sub-Hilbert space obtained by the action of the many-body projector, $\mathbb{P}$. The term $\mathbb{Q}H\mathbb{Q}$ acts purely outside the sub-Hilbert space spanned by $\mathbb{P}$. The off-diagonal term, $H_o = \mathbb{P}H\mathbb{Q} + \mathbb{Q}H\mathbb{P}$, mixes the low-energy states with the high-energy states.

Results

For a small A , we expand $H[A] = H[0] + J_\mu A_\mu + \frac{1}{2} K_{\mu\nu} A_\mu A_\nu + \dots \equiv H_d[A] + H_o[A]$, where J_μ and $K_{\mu\nu}$ are defined in Eq. 3. Clearly, $H_o[A]$ will introduce mixing between the active and remote bands. To leading order in $H_o[A]$, we can use the SW transformation to obtain a unitarily equivalent Hamiltonian, $\tilde{H}[A] = e^{T[A]} H[A] e^{-T[A]}$, which will have no matrix elements between the active and remote bands (30). The relevant matrix elements, $\langle m|T[A]|n\rangle = \langle m|H_o[A]|n\rangle / (E_m - E_n)$, where m, n correspond to energy levels of $H_d[A]$ with $|E_m - E_n| \geq \Delta$. This leads us to the $\mathcal{H}_{\text{eff}}[A] \equiv \mathbb{P}\tilde{H}[A]\mathbb{P}$, that was introduced in Eq. 1a:

$$\begin{aligned} \mathcal{H}_{\text{eff}}[A] &= \mathbb{P}H_d[A]\mathbb{P} \\ &+ \frac{1}{2} \sum_{\substack{m,n \in \mathbb{H}, \\ \ell \notin \mathbb{H}}} \left[\langle m|H_o[A]|\ell\rangle \langle \ell|H_o[A]|n\rangle \right. \\ &\times \left(\frac{1}{E_m - E_\ell} - \frac{1}{E_\ell - E_n} \right) \Big] + \dots, \end{aligned} \quad [7]$$

where we expand $\tilde{H}[A]$ using the Baker–Hausdorff–Campbell formula and keep only the leading order term. Higher-order terms are suppressed by either $1/\Delta$ or of a higher order than A^2 . We can now obtain the effective current, $J_{\text{x}}^{\text{eff}}$, and diamagnetic contribution, $K_{\text{xx}}^{\text{eff}}$, respectively, by expanding Eq. 7 up to second order in A and calculating the appropriate derivatives as in Eq. 1b. For $A = 0$, the first term in Eq. 7 ($= \mathbb{P}H\mathbb{P}$) is independent of Δ , while the second term is $O(V^2/\Delta)$. On the other hand, for $A \neq 0$, it is important to note that both the first and second terms in Eq. 7 contribute an $O(\Delta)$ correction to the $O(A^2)$ terms. Explicitly, the first term contributes $\mathbb{P}\hat{X}\mathbb{Q}H_{\text{kin}}\mathbb{Q}\hat{X}\mathbb{P} \sim O(\Delta)$, and the second term contributes $\mathbb{P}J_{\text{x}}\mathbb{Q}J_{\text{x}}\mathbb{P}/\Delta \sim O(\Delta)$ corrections, respectively. However, these $O(\Delta)$ contributions cancel when combined together and to the leading order $\langle K_{\text{xx}}^{\text{eff}} \rangle$ is independent of Δ . This cancellation is indeed expected since the result should be finite even in the limit $\Delta \rightarrow \infty$. The detailed derivation and explicit demonstration of the cancellation are discussed in *SI Appendix*.

In simplified form, the effective current operator becomes (after dropping a correction $\sim O(V^2/\Delta)$)

$$J_\mu^{\text{eff}}(\mathbf{q} \rightarrow 0) = \mathbb{P} \left(J_\mu(\mathbf{q} \rightarrow 0) + i \left[\hat{X}_\mu, H_o \right] \right) \mathbb{P}, \quad [8]$$

where the first term is just the projected microscopic current, while the second term arises from mixing between active and remote bands (31). Since H_{int} is a density–density interaction, $[\hat{X}_\mu, H_{\text{int}}] = 0$, and the second term in Eq. 8 can be rewritten as $-i \left[\mathbb{P} \hat{X}_\mu \mathbb{P}, \mathbb{P} H_{\text{int}} \mathbb{P} \right]$, which means that $J_\mu^{\text{eff}}(\mathbf{q} \rightarrow 0)$ can be expressed in terms of operators acting only within $\mathbb{H}$. Moreover, given the positive semidefinite property of $\chi_{\text{xx}}^{\text{eff}}$ (24), it is possible to express an upper bound on D_s in terms of the Δ –independent $\langle K_{\text{xx}}^{\text{eff}} \rangle$, which is exactly related to the diamagnetic response for the problem formulated directly in the projected limit (i.e., $\Delta \rightarrow \infty$).

The effective diamagnetic response is given by (upon retaining terms $\sim O(1)$),

$$\langle K_{\text{xx}}^{\text{eff}} \rangle = - \left\langle \left[\mathbb{P} \hat{X} \mathbb{P}, \left[\mathbb{P} \hat{X} \mathbb{P}, \mathbb{P} H_d \mathbb{P} \right] \right] \right\rangle. \quad [9]$$

The above result is already suggestive of an important subtlety, namely that it is only the projected degrees of freedom that enter $K_{\text{xx}}^{\text{eff}}$. This can be seen in a more transparent fashion by reexpressing it as:

$$\begin{aligned} \langle K_{\text{xx}}^{\text{eff}} \rangle &= \lim_{\alpha \rightarrow 0} \partial_\alpha^2 \langle e^{i\alpha \mathbb{P} \hat{X} \mathbb{P}} H_d e^{-i\alpha \mathbb{P} \hat{X} \mathbb{P}} \rangle \\ &= \langle K_{\text{xx}}^{\text{naive}} \rangle + \langle \hat{X} \mathbb{Q} \hat{X} \mathbb{P} H_d \mathbb{P} \rangle + \langle \mathbb{P} H_d \mathbb{P} \hat{X} \mathbb{Q} \hat{X} \rangle, \end{aligned} \quad [10a]$$

$$\text{where } \langle K_{\text{xx}}^{\text{naive}} \rangle \equiv \lim_{\alpha \rightarrow 0} \partial_\alpha^2 \langle e^{i\alpha \hat{X}} \mathbb{P} H_d \mathbb{P} e^{-i\alpha \hat{X}} \rangle. \quad [10b]$$

We can interpret Eq. 10a as an “effective” gauge transformation associated with the emergent conserved electronic charge density in the active bands, involving the projected position operator, $\mathbb{P} \hat{X} \mathbb{P}$. The action of $U_\alpha^{\text{eff}} \equiv e^{i\alpha \mathbb{P} \hat{X} \mathbb{P}} (= e^{i\alpha \sum_i x_i c_i^\dagger c_i})$ on any low-energy state belonging to $\mathbb{H}$ restricts it to the same Hilbert space, where the overline denotes projection to $\mathbb{H}$. The result of Eq. 10a is thus finite in the limit of $\Delta \rightarrow \infty$ and describes the intrinsic low-energy diamagnetic response associated with only the active bands. On the other hand, $\langle K_{\text{xx}}^{\text{naive}} \rangle$ in Eq. 10b represents the naive expectation for the diamagnetic response, where the usual gauge transformation $U_\alpha = e^{i\alpha \hat{X}}$ is carried out on the projected Hamiltonian. A priori, the latter procedure is not even guaranteed to lie solely within $\mathbb{H}$ and clearly differs from the procedure in Eqs. 9 and 10a.

It is worth noting that there is a “partial f-sum rule” for $T \ll \Delta_f$ that relates $\langle K_{\text{xx}}^{\text{eff}} \rangle$ to the integrated optical spectral weight associated with the interacting isolated bands, i.e.,

$$\int_0^\Lambda d\omega \text{Re}[\sigma_{\text{xx}}^{\text{eff}}(\omega)] = \frac{\pi e^2}{2} \langle K_{\text{xx}}^{\text{eff}} \rangle. \quad [11]$$

Here, the scale Λ is chosen with $\tilde{W} < \Lambda < \Delta$ (Fig. 1B) to obtain the integrated longitudinal optical conductivity for the interacting problem. Thus, the formalism developed in this paper also helps to address the related question of the contribution of the interacting isolated bands to the integrated optical spectral weight at low energies and with minor modifications can be

applied to other problems, e.g., magnetic circular dichroism (32–34).

To be explicit, performing the usual gauge transformation generated by U_α on the active degrees of freedom (e.g., assuming that there is only one active band) and then projecting back is equivalent to the operation, $c_{\mathbf{k}} \rightarrow \mathbb{P} U_\alpha^\dagger c_{\mathbf{k}} U_\alpha \mathbb{P} = c_{\mathbf{k}+\alpha \mathbf{e}_x} \langle u_{\mathbf{k}} | u_{\mathbf{k}+\alpha \mathbf{e}_x} \rangle$, where $\mathbf{e}_x$ is the unit vector along x direction. This restricted transformation is nonunitary, modifying the measure of the path integral as $|\langle u_{\mathbf{k}} | u_{\mathbf{k}+\alpha \mathbf{e}_x} \rangle|^2 \approx 1 - \alpha^2 g_{\text{xx}}(\mathbf{k})$, where $g_{\text{xx}}(\mathbf{k})$ is the quantum metric. This issue of nonunitarity is manifest in some properties of $\langle K_{\text{xx}}^{\text{naive}} \rangle$. First of all, $\langle K_{\text{xx}}^{\text{naive}} \rangle$ is not intrinsic to solely the active degrees of freedom but instead also depends on the remote bands in the following sense. If we shift the energy levels of the active bands by a constant amount, say $\mathbb{P} H_{\text{kin}} \mathbb{P} \rightarrow \mathbb{P} H_{\text{kin}} \mathbb{P} + \mathbb{P} \delta$, we have $\langle K_{\text{xx}}^{\text{naive}} \rangle \rightarrow \langle K_{\text{xx}}^{\text{naive}} \rangle - \delta \sum_{\mathbf{k}} 2n_{\mathbf{k}} g_{\text{xx}}(\mathbf{k})$. Moreover, if the active bands are fully filled, the resulting insulator should have a vanishing $\langle K_{\text{xx}}^{\text{eff}} \rangle$; however, $\langle K_{\text{xx}}^{\text{naive}} \rangle$ does not necessarily vanish. To see this dichotomy, note that $e^{i\alpha \mathbb{P} \hat{X} \mathbb{P}}$ is a unitary operator acting on $\mathbb{H}$ and does not change the particle numbers of the active bands. Therefore, if $|\psi\rangle$ is a state with fully filled active bands, $e^{i\alpha \mathbb{P} \hat{X} \mathbb{P}} |\psi\rangle = e^{i\theta_\alpha} |\psi\rangle$, where θ_α is a phase factor. As a result, $\langle \psi | e^{-i\alpha \mathbb{P} \hat{X} \mathbb{P}} H_d e^{i\alpha \mathbb{P} \hat{X} \mathbb{P}} | \psi \rangle = \langle \psi | H_d | \psi \rangle$, and $\langle K_{\text{xx}}^{\text{eff}} \rangle = 0$. On the other hand, in the same state, we obtain $\langle K_{\text{xx}}^{\text{naive}} \rangle = 2E_0 \sum_{\mathbf{k}, m} g_{\text{xx}}^{mm}(\mathbf{k})$, which is in general nonzero. Here, E_0 is the energy of the insulating state $|\psi\rangle$, and $g_{\text{xx}}^{mm}(\mathbf{k})$ is the quantum metric generalized to multiple orbitals, defined in [SI Appendix](#).

Estimates for Microscopic Models. We can now turn to estimating the different intrinsic contributions from the active bands to $\langle K_{\text{xx}}^{\text{eff}} \rangle$ in order to place an upper bound on $D_s \leq \pi e^2 \langle K_{\text{xx}}^{\text{eff}} \rangle$. Specifically, there are three distinct contributions from the active bands to D_s that originate from i) the bare dispersion, ii) the interaction-induced dispersion, and iii) the diamagnetic response due to projected interactions. The first two can be combined together to yield:

$$\left. \langle K_{\text{xx}}^{\text{eff}} \rangle \right|_{\text{kinetic}} = \sum_{\mathbf{k}} \frac{\partial^2 \tilde{\varepsilon}_{\mathbf{k}m}}{\partial k_x^2} \langle c_{\mathbf{k}m}^\dagger c_{\mathbf{k}m} \rangle, \quad \text{where} \quad [12a]$$

$$\tilde{\varepsilon}_{\mathbf{k}} = \varepsilon_{\mathbf{k}m} + \sum_{\mathbf{q}} V(\mathbf{q}) |\langle u_{\mathbf{k},m} | u_{\mathbf{k}-\mathbf{q},m} \rangle|^2, \quad [12b]$$

and we have focused on the situation with only one active band (i.e., m is a fixed label) for simplicity. The more general expression appears in [SI Appendix](#).

The purely interaction-induced contribution can be expressed as:

$$\left. \langle K_{\text{xx}}^{\text{eff}} \rangle \right|_{\text{int}} = \sum_{\mathbf{k}_1, \mathbf{k}_2, \mathbf{q}} V(\mathbf{q}) F(\mathbf{k}_1, \mathbf{k}_2, \mathbf{q}) \langle c_{\mathbf{k}_1 m}^\dagger c_{\mathbf{k}_2 m}^\dagger c_{\mathbf{k}_2 + \mathbf{q} m'} c_{\mathbf{k}_1 - \mathbf{q} m'} \rangle, \quad [13]$$

where $F(\mathbf{k}_1, \mathbf{k}_2, \mathbf{q})$ can be expressed as:

$$F(\mathbf{k}_1, \mathbf{k}_2, \mathbf{q}) = \left[\hat{D}_{\mathbf{k}_1}^x + \hat{D}_{\mathbf{k}_2}^x \right]^2 \langle u_{\mathbf{k}_1, m} | u_{\mathbf{k}_1 - \mathbf{q}, m} \rangle \langle u_{\mathbf{k}_2, m'} | u_{\mathbf{k}_2 + \mathbf{q}, m'} \rangle. \quad [14]$$

Here, $\hat{\mathcal{D}}_{\mathbf{k}}^\mu$ can be viewed as a covariant derivative in $\mathbf{k}$ -space acting on the form factor as:

$$\begin{aligned} & [\hat{\mathcal{D}}_{\mathbf{k}}^\mu]_{mm',nn'} \langle u_{\mathbf{k},m'} | u_{\mathbf{k}-\mathbf{q},n'} \rangle \\ &= (\partial_{k_\mu} \delta_{mm'} \delta_{nn'} - i \mathcal{A}_{\mathbf{k},mm'}^\mu \delta_{nn'} + i \mathcal{A}_{\mathbf{k}-\mathbf{q},n'n}^\mu \delta_{mm'}) \langle u_{\mathbf{k},m'} | u_{\mathbf{k}-\mathbf{q},n'} \rangle, \end{aligned} \quad [15]$$

with $\mathcal{A}_{\mathbf{k},mm'}^\mu = i \langle u_{\mathbf{k},m} | \partial_{k_\mu} u_{\mathbf{k},m'} \rangle$ the multiorbital Berry connection. We note that given $\mathbf{k}_1$, $\mathbf{k}_2$, $\mathbf{q}$ are unrelated to each other, for a generic $V(\mathbf{q})$, the function $F(\dots)$ hosts seemingly nonlocal correlations in momentum space (35). In order to proceed with Eq. 13 further, one option is to evaluate the expectation value in a specific many-body state for a given Hamiltonian. In general, this is difficult as obtaining the true many-body correlations in the state of interest is a challenging affair. An alternative approach is to make approximations for the four-point expectation values in terms of the band-filling (36) and replace each term in the sum by its magnitude to obtain a conservative upper bound on $\langle K_{xx}^{\text{eff}} \rangle_{\text{int}}$ and T_c .

Topologically Trivial Flat-Band Model. One of the most transparent and straightforward applications of our formalism is to the problem of interacting isolated bands that are topologically trivial. Here, we can express $\mathbb{P}H_d\mathbb{P}$ in terms of a basis of exponentially localized Wannier states and obtain the effective electromagnetic response for the interacting theory in the projected Hilbert space. Note that the “wannierization” of the underlying Bloch states is not an essential element of our framework. To highlight the key insight, we demonstrate the procedure to evaluate $\langle K_{xx}^{\text{eff}} \rangle$ in the situation with just one active (but possible degenerate) set of bands. Here, we can always choose a gauge where the Berry connection $\mathcal{A}_k \equiv -i \langle u_{\mathbf{k}} | \partial_{\mathbf{k}} u_{\mathbf{k}} \rangle = 0$. The effective contribution is then determined by

$$\begin{aligned} \langle K_{xx}^{\text{eff}} \rangle &= \sum_{\mathbf{k},\alpha} \partial_{k_x}^2 \varepsilon_{\mathbf{k},\alpha} \langle c_{\mathbf{k},\alpha}^\dagger c_{\mathbf{k},\alpha} \rangle \\ &+ \sum_{\substack{\mathbf{q},\mathbf{k}_1,\mathbf{k}_2 \\ \alpha,\beta}} \Xi(\mathbf{k}_1, \mathbf{k}_2, \mathbf{q}) \langle c_{\mathbf{k}_1,\alpha}^\dagger c_{\mathbf{k}_1-\mathbf{q},\alpha} c_{\mathbf{k}_2,\beta}^\dagger c_{\mathbf{k}_2+\mathbf{q},\beta} \rangle, \end{aligned} \quad [16a]$$

$$\Xi(\mathbf{k}_1, \mathbf{k}_2, \mathbf{q}) = V(\mathbf{q}) (\partial_{k_{1,x}} + \partial_{k_{2,x}})^2 g_{\alpha\beta}(\mathbf{k}_1, \mathbf{k}_2, \mathbf{q}), \quad [16b]$$

where $g_{\alpha\beta}(\mathbf{k}_1, \mathbf{k}_2, \mathbf{q}) = \langle u_{\mathbf{k}_1,\alpha} | u_{\mathbf{k}_1-\mathbf{q},\alpha} \rangle \langle u_{\mathbf{k}_2,\beta} | u_{\mathbf{k}_2+\mathbf{q},\beta} \rangle$ and α, β label the spin indices. In terms of the operators d_i , $d_i^\dagger$ corresponding to the Wannier orbitals centered at a site $\mathbf{r}_i$ spanning the active bands,

$$\begin{aligned} \langle K_{xx}^{\text{eff}} \rangle &= \sum_{i,j} \langle d_{i,\alpha}^\dagger d_{j,\alpha} \rangle \overbrace{\sum_{\mathbf{k}} e^{i\mathbf{k} \cdot (\mathbf{r}_i - \mathbf{r}_j)} \partial_{k_x}^2 \varepsilon_{\mathbf{k},\alpha}}^{\mathcal{X}} \\ &+ \sum_{\substack{i,j,l,m \\ \alpha,\beta}} \langle d_{i,\alpha}^\dagger d_{j,\alpha} d_{l,\beta}^\dagger d_{m,\beta} \rangle \\ &\times \underbrace{\sum_{\mathbf{q},\mathbf{k}_1,\mathbf{k}_2} \Xi(\mathbf{k}_1, \mathbf{k}_2, \mathbf{q}) e^{i\mathbf{k}_1 \cdot (\mathbf{r}_i - \mathbf{r}_j) + i\mathbf{k}_2 \cdot (\mathbf{r}_l - \mathbf{r}_m) - i\mathbf{q} \cdot (\mathbf{r}_j - \mathbf{r}_m)}}_{\mathcal{Y}}. \end{aligned} \quad [17]$$

The coefficients $\mathcal{X}$, $\mathcal{Y}$ can be reexpressed in a more transparent fashion as:

$$\mathcal{X} = -(x_i - x_j)^2 \sum_{\mathbf{k}} e^{i\mathbf{k} \cdot (\mathbf{r}_i - \mathbf{r}_j)} \varepsilon_{\mathbf{k},\alpha} \quad [18a]$$

$$\begin{aligned} \mathcal{Y} &= -(x_i - x_j + x_l - x_m)^2 \\ &\times \sum_{\mathbf{q},\mathbf{k}_1,\mathbf{k}_2} V(\mathbf{q}) e^{i\mathbf{k}_1 \cdot (\mathbf{r}_i - \mathbf{r}_j) + i\mathbf{k}_2 \cdot (\mathbf{r}_l - \mathbf{r}_m) - i\mathbf{q} \cdot (\mathbf{r}_j - \mathbf{r}_m)} g_{\alpha\beta}(\mathbf{k}_1, \mathbf{k}_2, \mathbf{q}). \end{aligned} \quad [18b]$$

Therefore, for topologically trivial bands, the coefficients $\mathcal{X}$, $\mathcal{Y}$ can be obtained from the low-energy projected Hamiltonian expressed in terms of Wannier orbitals by carrying out the usual Peierls’ substitution, $d_{i,\alpha} \rightarrow d_{i,\alpha} e^{i\mathbf{A} \cdot \mathbf{r}_i}$, followed by $\delta^2 \mathcal{H}_{\text{eff}}[A]/\delta A^2$. However, importantly, this procedure is only valid for maximally localized Wannier orbitals (19).

It is useful to focus on a concrete model, where we can compare our approximate estimate of $D_s^{\text{upper}} = \pi e^2 \langle K_{xx}^{\text{eff}} \rangle$ with the known D_s evaluated, e.g., using numerically exact quantum Monte-Carlo (QMC) computations. With this in mind, we focus on a two-orbital, spinful time-reversal symmetric model of flat bands with a tunable quantum metric (25). In the presence of local competing interactions, sign-problem-free QMC calculations have revealed a number of intertwined orders (23). The Hamiltonian is given by $H = H_{\text{kin}} + H_{\text{int}}$,

$$H_{\text{kin}} = -t \sum_{\mathbf{k}} \hat{\mathbf{c}}_{\mathbf{k}}^\dagger (\tau_x \sin \alpha_{\mathbf{k}} + \sigma_z \tau_y \cos \alpha_{\mathbf{k}} + \mu \tau_0) \hat{\mathbf{c}}_{\mathbf{k}}, \quad \text{where} \quad [19a]$$

$$H_{\text{int}} = -\frac{U}{2} \sum_{\mathbf{r},l} \delta \hat{n}_{\mathbf{r},l}^2 + V \sum_{\langle \mathbf{r},\mathbf{r}' \rangle, l} \delta \hat{n}_{\mathbf{r},l} \delta \hat{n}_{\mathbf{r}',l}. \quad [19b]$$

Here, $\hat{\mathbf{c}}_{\mathbf{k}}^\dagger$ represents the electron creation operators with momentum $\mathbf{k}$ and additional labels—spin $s = \uparrow, \downarrow$ and orbital $l = 1, 2$. The Pauli-matrices σ_j and τ_j act on the spin and orbital indices, respectively. We set the lattice constant, $a = 1$ henceforth. The model is engineered to yield two flat bands with $\varepsilon_{\mathbf{k}} = \pm t$ irrespective of the values of t and the parameter ζ ; the latter directly controls the minimal spatial extent associated with the exponentially localized Wannier functions. It is also worth noting that for the present model, the Berry curvature vanishes identically everywhere in the Brillouin zone, while the Fubini–Study metric is finite and integrates to $\zeta^2/2$. Finally, H_{int} includes an on-site attraction ($U > 0$) and a nearest-neighbor interaction with $\delta \hat{n}_{\mathbf{r},l} = \sum_s \hat{c}_{\mathbf{r},l,s}^\dagger \hat{c}_{\mathbf{r},l,s} - 1$.

For this model, we can evaluate $\langle K_{xx}^{\text{eff}} \rangle$ for the active spin-degenerate flat bands by choosing the gauge where $\mathcal{A}_k \equiv -i \langle u_{\mathbf{k}} | \partial_{\mathbf{k}} u_{\mathbf{k}} \rangle = 0$, and with $V(\mathbf{q}) = U + V[\cos q_x + \cos q_y]$. To first gain intuition into what the expansion in Eq. 17 represents, we can Fourier-transform these terms back to real space. Specifically, these include contributions to $\langle K_{xx}^{\text{eff}} \rangle$ from an interaction-mediated hopping term, $\langle K_{xx}^{\text{eff}} \rangle_{\text{hop}}$, and a pair-hopping term, $\langle K_{xx}^{\text{eff}} \rangle_{\text{pair}}$, respectively,

$$\langle K_{xx}^{\text{eff}} \rangle_{\text{hop}} \sim \sum_{i,a_1,a_2,\sigma} (x_{a_1} - x_{a_2})^2 \langle d_{i,\uparrow}^\dagger d_{i,\uparrow} d_{i+a_1,\sigma}^\dagger d_{i+a_2,\sigma} \rangle, \quad [20a]$$

$$\langle K_{xx}^{\text{eff}} \rangle_{\text{pair}} \sim \sum_{\langle i,j \rangle} (2x_i - 2x_j)^2 \langle d_{i,\uparrow}^\dagger d_{i,\downarrow}^\dagger d_{j,\downarrow} d_{j,\uparrow} \rangle. \quad [20b]$$

The above exercise also makes it clear that the density–density interactions that are generated within $\mathcal{H}_{\text{eff}}$ (originating from the

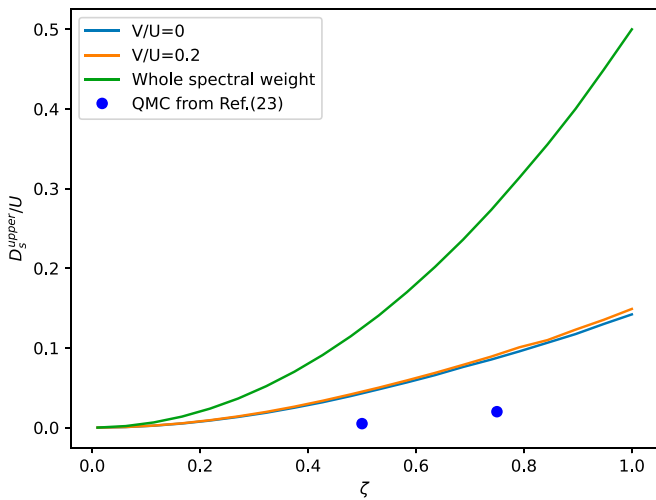


Fig. 3. Estimate of D_s^{upper} for the interacting flat-band model in Eq. 19b evaluated at $T = 0$ and for quarter-filling as a function of ζ . The QMC results for $D_s(0)$ are taken from Ref. (23) for $V = 0$. The whole spectral weight (green curve) is set by the total bandwidth t and includes contributions from both flat bands at $\epsilon_{\mathbf{k}} = \pm t$. The optical spectral weight for the interaction projected to the lower flat band, $\epsilon_{\mathbf{k}} = -t$, is shown for two different values of V (blue and orange curves). We have taken $t = 4U$ for comparison of the low-energy bound with the bound based on whole spectral weight.

V -term) do not contribute to K_{xx}^{eff} as these do not couple to the vector potential.

Finally, we turn to an explicit numerical evaluation of D_s^{upper} at $T = 0$ directly in momentum space and a comparison with the exact results obtained using QMC (23). Note that for this flat-band model, there is no bound on T_c/E_F with an appropriately defined E_F (25). We plot D_s^{upper} with respect to ζ in Fig. 3. First, we note that in the limit of $\zeta \rightarrow 0$, the sites are disconnected completely such that the optical spectral weight (and relatedly the diamagnetic response) vanishes identically. We can first evaluate the upper bound as determined by the total spectral weight (green line in Fig. 3), which includes both bands at $\epsilon_{\mathbf{k}} = \pm t$ and the interband matrix-elements. For small ζ , this spectral weight scales as $\sim \zeta^2 t$. Turning to the upper bound as determined by H_{int} projected to the lower flat band, the effective D_s^{upper} is much closer in magnitude to the result from QMC (23). It is interesting to note that for a finite V , D_s^{upper} is modified, in part due to contributions from a term such as $\langle K_{xx}^{\text{eff}} \rangle|_{\text{hop}}$. However, in the actual QMC computations, the ground state is also more susceptible to forming a charge-density wave with increasing V (23); the finite result for D_s^{upper} guarantees an integrated optical

spectral weight via the “partial f-sum rule” but does not guarantee an actual superconducting ground state.

We end by noting that recent work (36) has carried out a similar analysis on models with local on-site interactions projected to flat bands and obtained the associated pair-hopping terms for a “wannierized” description. In general, for an arbitrary gauge choice, or for topological bands where there is an obstruction to constructing exponentially localized Wannier orbitals, one has to use the full gauge-invariant expression for K_{xx}^{eff} .

Outlook

The central goal of this paper has been to derive the effective electromagnetic response function vis-à-vis the phase stiffness (or, the integrated optical spectral weight) in terms of only the low-energy degrees of freedom in the active isolated bands of interest. The formulas derived here can be combined with sophisticated numerical methods to evaluate these quantities accurately. Given the generality of the formalism, it can be applied to a wide range of interacting Hamiltonians, including lattice interacting tight-binding models, and momentum-space models. Similar methods can also be applied to address bounds on the spin stiffness for interacting ferromagnets in flat-band systems. A number of recent theoretical works have studied the interplay of berry curvature distribution and quantum geometry in interacting topological flat bands on the stability of fractional Chern insulators (37–40). Finding the fundamental ingredients that guarantee a robust superconducting ground state remains an interesting problem for the future. We end by noting that we have derived the diamagnetic response from the effective Hamiltonian, $\mathcal{H}_{\text{eff}}[A]$, in the presence of a constant (probe) vector potential A . The more general response functions in the low-energy effective theory for a spacetime-dependent vector potential have a rich structure and are left for future work.

Data, Materials, and Software Availability. There are no data underlying this work.

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